

3.1.6 Modern industrial and commercial practice

Scales of production

Content	Potential links to maths and science
Students should be aware of, and be able to describe, the different scales of production giving example products and specific manufacturing methods. Specific scales of production to include: • one-off, bespoke • batch production • mass/line production • unit production systems (UPS) • quick response manufacturing (QRM) • vertical in-house production.	

3.1.6.2 Efficient use of materials

Content	Potential links to maths and science
Students must develop an awareness of the relationship between material cost, form, and manufacturing processes, and the scale of production. • The development of designs which use materials economically and with regard to their characteristics. • The use of manufacturing processes which increase accuracy and reduce waste. • The savings to be gained when comparing bulk production with one-off production. • The advantages of Just In Time (JIT) manufacture.	Determining quantities of materials.

The use of computer systems

Content	Potential links to maths and science
Students should be aware of how computer systems are used to plan and control manufacturing, reduce waste and respond quickly to changes in consumer demand.	
Students should be able to explain specific industrial manufacturing systems and their use in the production of given products. Specific manufacturing systems to include: • modular/cell production • just in time (JIT) • quick response manufacturing (QRM) • flexible manufacturing systems.	
Students should be able to explain the use of computer controlled systems in production, distribution and storage. Students should be able to explain the use of standardised and bought-in components made by specialist manufacturers.	

Sub-assembly

Content	Potential links to maths and science
Students should be aware of, and able to explain, sub-assembly as a separate line of manufacture for certain parts of a product.	